8.2.3 Three point starter

Three point starters are used to start the shunt motor. The diagram of the three point starter is shown in figure 8.1. In three point starter, three terminals L, A and F (Line, Armature and Field) are available. In this starter, the resistance elements are mounted on the back side of a slate board. On the front side of the board, brass studs are provided and the resistance junctions are connected with each brass stud. Two protective devices overload release and no volt release is incorporated in the circuit as shown in the diagram. The handle of the starter is fixed in such a way to move the brass studs.

When the handle touches the first stud, the full resistance is connected in series with the armature. The field circuit is connected across the full supply voltage. The handle is moved over the studs, the resistance connected in series with the armature circuit is gradually cut off. The handle movement is controlled by the tension of the spring as shown in the figure 8.1.

A soft iron piece is attached to the handle. The soft iron piece is attached by the electro-magnet (NVR) when the handle reaches the 'ON' position. In case of a failure of the supply or the voltage is very low, the electro magnet de-energizes and releases the armature. The spring forces bring the handle to 'OFF' position.

Protective devices

No-Volt Release (NVR)

NVR consists of an electromagnet. It is connected in the field circuit. As soon as the field circuit gets supply, this is energized. It holds the handle in the 'ON' position. As explained above, in case of a failure of the supply, this de-energizes and the handle is released from 'ON' position. The handle returns to 'OFF' position, due to the action of the spring. If this provision is not provided, the supply restores, the current through the armature becomes high and this will damage the armature windings.

Over load release (OLR)

OLR is also consists of an electro magnet. This electromagnet coil is energized by the line current. When the load on the motor is increased above pre-determined value, the magnetizing force established is sufficient to lift the movable iron. When it is lifted by the electro magnet of OLR, it short circuits the terminals of the coils of the no volt release. Hence, the no volt coil is de-energized and the starter handle returns to 'OFF' position. Thus the overload release protects the motor against overloads.

This starter is not generally used where the field current is often adjusted for a higher speed than the normal speed, the handle returns to the 'OFF' position. This may cause for reduction in field current.

Disadvantages

The 3 point starter suffers from a serious drawback for motors with a large variation of speed by adjustment of the field rheostat. To increase the speed of the motor, the field resistance should be increased. Therefore, the current through

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